

1)Go Installation

- Install Go and verify the installation using `go version`.
- Run `go env` and identify `GOROOT` and `GOPATH`.

```
PS C:\Users\M.Chanikya\Downloads\Overture_Project\data\GO\assignment> go version
go version go1.27.1 windows/amd64
PS C:\Users\M.Chanikya\Downloads\Overture_Project\data\GO\assignment> go env

set GOPATH=C:\Users\M.Chanikya\go
set GOPRIVATE=
set GOPROXY=https://proxy.golang.org,direct
set GOROOT=C:\Program Files\Go
```

4)Go Toolchain Practice

- Execute:
 - `go run`
 - `go build`
 - `go fmt`
 - `go vet`
 - `go test`
- Document the purpose of each command.

```
PS C:\Users\M.Chanikya\Downloads\Overture_Project\data\GO\assignment> go run toolchain.go
toolchain
PS C:\Users\M.Chanikya\Downloads\Overture_Project\data\GO\assignment> go build toolchain.go
PS C:\Users\M.Chanikya\Downloads\Overture_Project\data\GO\assignment> go fmt toolchain.go
toolchain.go
PS C:\Users\M.Chanikya\Downloads\Overture_Project\data\GO\assignment> go vet toolchain.go
PS C:\Users\M.Chanikya\Downloads\Overture_Project\data\GO\assignment> go test toolchain.go
?      command-line-arguments  [no test files]
```

go run: Compiles and runs your Go program instantly without saving a permanent executable file.

go build: Compiles your Go source code and generates a reusable, standalone executable binary file.

go fmt: Automatically formats your Go source code files to match the official Go style and indentation standards.

go vet: Scans your Go source code for suspicious constructs, structural mistakes, and common logic bugs before runtime.

go test: Automatically discovers and executes any unit test functions written in your project's `_test.go` files.

6 External Package

- Add an external Go dependency.
- Use it in a meaningful example.

```
PS C:\Users\M.Chanikya\Downloads\Overture_Project\data\GO\assignment> go get github.com/google/uuid
go: added github.com/google/uuid v1.6.0
PS C:\Users\M.Chanikya\Downloads\Overture_Project\data\GO\assignment> go run example.go
Generated UUID: e8cd1875-73bd-4fca-9d04-e8c2813ee942
```